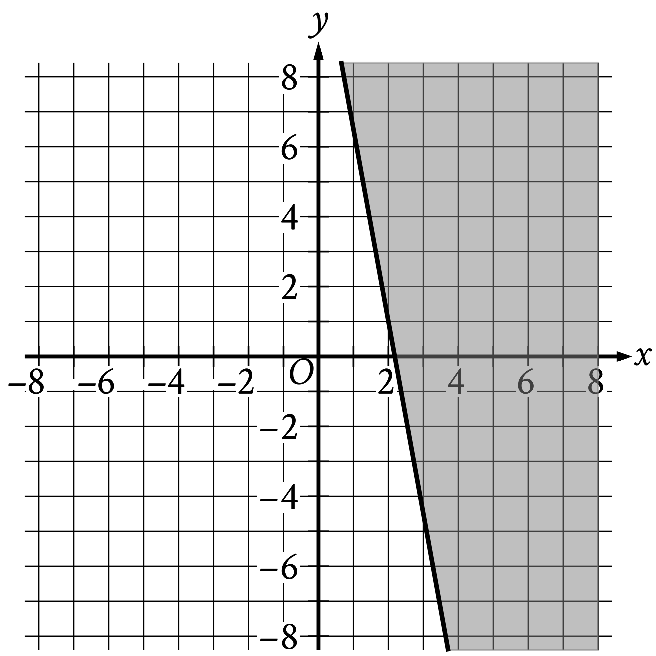
EASY ALGEBRA

# Algebra

08

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29




- The boundary of the inequality is a solid line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (1.0 comma 6.5)
    - (3.0 comma negative 4.5)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents solutions to an inequality. Which ordered pair $x, y$ is a solution to this inequality?

**A.** $0, -4$

**B.** $0, 4$

**C.** $-4, 0$

**D.** $4, 0$

**Q2** ALGEBRA

An employee at a restaurant prepares sandwiches and salads. It takes the employee $1.5$ minutes to prepare a sandwich and $1.9$ minutes to prepare a salad. The employee spends a total of $46.1$ minutes preparing $x$ sandwiches and $y$ salads. Which equation represents this situation?

**A.** $1.9x + 1.5y = 46.1$

**B.** $1.5x + 1.9y = 46.1$

**C.** $x + y = 46.1$

**D.** $30.7x + 24.3y = 46.1$

**Q3** ALGEBRA

What is the solution to the equation $2x + 3 = 7$?

**A.** $1$

**B.** $1.5$

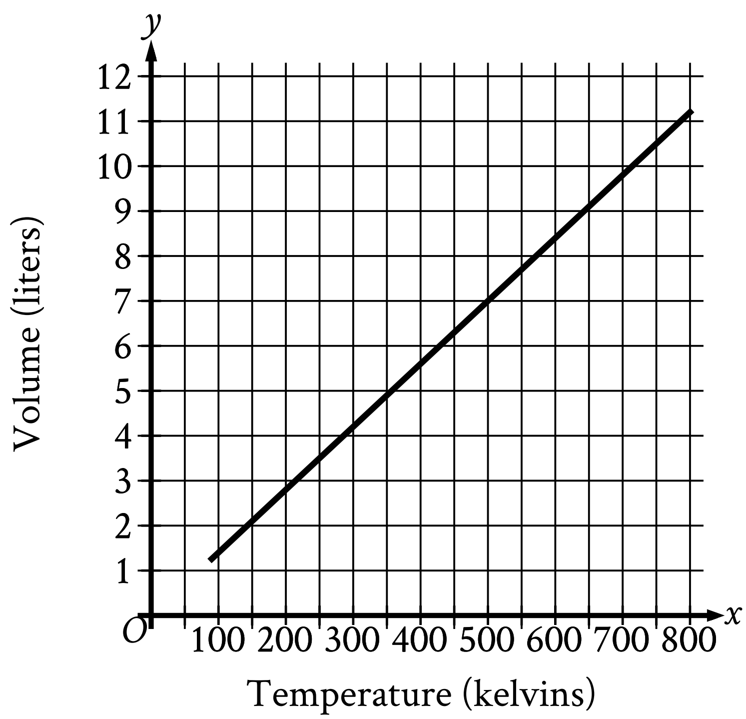
**C.** $2$

**D.** $4$

Q4

ALGEBRA

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume $y$, in liters, of the hydrogen when its temperature is $x$ kelvins.




- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.3).
- The line passes through the following approximate points:
  - (90 comma 1.3)
  - (500 comma 7)

What is the estimated volume, in liters, of the hydrogen when its temperature is $500$ kelvins?

**A.** $0$

**B.** $\frac{7}{500}$

C. 7

D. $\frac{500}{7}$

Q5 ALGEBRA

$$3x = 12$$

$$-3x + y = -6$$

The solution to the given system of equations is $x, y$. What is the value of $y$?

A. -3

B. 6

C. 18

D. 30

Q6 ALGEBRA

The point $(8, 2)$ in the $xy$-plane is a solution to which of the following systems of inequalities?

A. $x > 0$
$y > 0$

B. $x > 0$
$y < 0$

C. $x < 0$
$y > 0$

D. $x < 0$
$y < 0$

Q7

ALGEBRA

$$x + y = 75$$

The equation above relates the number of minutes, $x$, Maria spends running each day and the number of minutes, $y$, she spends biking each day. In the equation, what does the number 75 represent?

A. The number of minutes spent running each day

B. The number of minutes spent biking each day

C. The total number of minutes spent running and biking each day

D. The number of minutes spent biking for each minute spent running

Q8

ALGEBRA

$$\frac{4x}{5} = 20$$

In the equation above, what is the value of x ?

A. 25

B. 24

C. 16

D. 15

Q9

ALGEBRA

The function $f$ is defined by $fx = \frac{1}{10}x$ - 2. What is the $y$-intercept of the graph of $y = fx$ in the $xy$-plane?

A. -2, 0

B. 0, -2

C. 0, $\frac{1}{10}$

D. $\frac{1}{10}$, 0

Q10

ALGEBRA

$$4x = 20$$

$$-3x + y = -7$$

The solution to the given system of equations is $x, y$. What is the value of $x + y$?

A. -27

B. -13

C. 13

D. 27